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The Dangers of Human Touch: Fingerprinting Browser Extensions through User Actions

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Browser extension fingerprinting has garnered considerable attention recently due to the twofold privacy loss that it incurs. Apart from facilitating tracking by augmenting browser fingerprints, the list of installed extensions can be directly used to infer sensitive user characteristics. However, prior research was performed in a vacuum, overlooking a core dimension of extensions' functionality: how they react to user actions. In this paper, we present the first exploration of user-triggered extension fingerprinting. Guided by our findings from a large-scale static analysis of browser extensions we devise a series of user action templates that enable dynamic extension-exercising frameworks to comprehensively uncover hidden extension functionality that can only be triggered through user interactions. Our experimental evaluation demonstrates the effectiveness of our proposed technique, as we are able to fingerprint 4,971 unique extensions, 36% of which are not detectable by state-of-the-art techniques. To make matters worse, we find that $\approx 67\%$ of the extensions that require mouse or keyboard interactions lack appropriate safeguards, rendering them vulnerable to pages that simulate user actions through JavaScript. To assist extension developers in protecting users from this privacy threat, we build a tool that automatically includes origin checks for fortifying extensions against invasive sites.

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